

# B D S Aritra

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## EXPERIENCE

### Jefferies

Jul 2025 – Present

*AI Software Engineer, Operations Technology*

*Jersey City, NJ*

#### AI Trade Confirmation Reconciliation Platform | *Python, TypeScript, React, Node.js, MCP, PostgreSQL*

- Sole engineer owning product, build and DevOps; shipped in 3 months and adopted by **1,000+ users** across all trading middle-office teams, cutting per-trade review from **15–45 min to under 60 s**
- Hit a **98% straight-through-processing rate** (85% target) across 7 asset classes and 43 trade types, routing each PDF/Excel confirmation via an LLM classifier to versioned prompts
- Architected a React/TypeScript UI, Node.js (Hono) gateway with Azure AD auth, RBAC and streaming LLM orchestration, and 7 isolated Python FastMCP servers running up to 25 tool calls per trade
- Built an eval harness with per-trade-type assertion validators to regression-test prompt changes, plus full audit logging (prompt, model, tool calls, tokens) and an STP analytics dashboard

#### Agentic Operations Platform for Back-Office Teams | *TypeScript, React, PostgreSQL, Oracle, AWS Bedrock*

- As sole engineer, built a unified platform hosting apps for **5 operations teams** behind one API gateway, Azure AD SSO and team-scoped authorization, replacing spreadsheets and legacy terminals
- Built a Claude agent (AWS Bedrock, 25-step tool loop) resolving **nearly 100K unmapped counterparty-code exceptions** via Oracle queries, Kerberos-secured APIs and email search, with analyst sign-off
- Building a daily reconciliation of DTCC swap-repository reports against 4 trading systems (FX, rates, equities, credit) to flag alleged and orphan trades under CFTC, SEC and CSA rules
- Automated monthly TBA settlement-date entry into a third-party settlement system; leading rollout to client onboarding for document-collection and sanctions-screening agents

### Jefferies

Jun 2024 – Aug 2024

*Corporate Technology Intern*

*Jersey City, NJ*

- Built a full-stack financial operations dashboard (Java, Spring Batch, Angular, AG Grid), cutting Oracle load time from **12 s to under 1 s** through indexing and server-side pagination
- Added Cypress end-to-end tests covering 90%+ of critical flows in CI/CD (Bamboo), cutting regression bugs **40%**

### Trinity College

Jan 2023 – May 2025

*Software Engineer & ML Research Assistant*

*Hartford, CT*

- Shipped DiscountBytes (React, Firebase, Tailwind CSS), a group-discount app with occupancy-based dynamic pricing and real-time coupons; signed 3 partner restaurants, lifted partner traffic **20%**, held 98% uptime
- Trained CNN, GAN and LSTM models (TensorFlow, Keras) on NASA Solar Dynamics Observatory imagery; built a LangChain pipeline that cut training-to-inference time **30%** and improved accuracy **15%**

## EDUCATION

### Trinity College

Hartford, CT

*B.S. with Honors, Mathematics & Computer Science; GPA: 3.97/4.00*

*Aug 2021 – May 2025*

- Teaching Assistant for 5 courses, including Data Structures & Algorithms, Computer Systems and Discrete Math

## PUBLICATIONS

*Scalability Matters: Overcoming Challenges in InstructGLM with Similarity-Degree-Based Sampling*

IJCNN 2025

*Predicting Coronal Hole Activity: Key to Mitigating Space Weather Impacts*

PAAISS 2024

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, Java, C

**AI / LLM:** Agentic systems, tool calling, MCP, LLM evals, prompt engineering, Vercel AI SDK, LangChain

**ML & Data:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, Pandas, NumPy, PySpark

**Backend:** Node.js, Hono, Bun, Spring Boot, Flask, REST APIs, PostgreSQL, Prisma, Oracle, MySQL, MongoDB

**Frontend:** React, Vite, Tailwind CSS, Angular, AG Grid

**Cloud & DevOps:** AWS (Bedrock, Lambda, S3, EKS), Azure AD, Docker, Kubernetes, Jenkins, CI/CD, GCP